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Experimental study of a metal hydride vessel based on a finned spiral heat exchanger

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ABSTRACT

The reaction time of hydrogen in metal hydride vessels (MHVs for short) is strongly influenced by the heat transfer from/to the hydride bed. In the present work an experimental study of the geometric and the operating parameters of a finned spiral heat exchanger has been carried out to identify their influence on the performance of the charging process of the MHV. The experimental results show that the charge time of the reactor is considerably reduced, when finned spiral heat exchanger is used. In addition, the effect of different parameters (flow mass and temperature of the cooling fluid, applied pressure, and hydrogen tank volume) has been discussed and obtained results show that a good choice of these parameters is important.

1. Introduction

Hydrogen absorbing alloys can be applied for industrial applications that have considerable potential, such as hydrogen storage, chemical compressors and chemical heat pumps. The main features required for these alloys are generation of a large amount of heat during absorption, high volumetric density and reversibility [1].

The dynamic response of hydrogen storage system is important in the design of metal hydride energy conversion systems. However, it is known that the performance of hydrogen absorption and desorption in the case of metal hydrides depends on heat and mass transfer within the metal hydride beds. Hence, for the design of an optimized metal hydride vessel (MHV), it is necessary to understand and predict transient heat and mass transfer into the metal hydride bed. Therefore, there have been numerous experimental and theoretical investigations on several aspects of hydrogen storage. Gopal et al. [2,3] and Mayer et al. [4] investigated transient heat and mass transfer within the reactor using a one-dimensional mathematical model. Jemni and Ben Nasrallah [5,6] examined two dimensional heat and mass transfer within a metal-hydrogen reactor for both absorption and desorption processes. The sorption process for mixtures containing impurities that are inactive or weakly active to the hydride forming material was investigated by Shmalkov et al. [7]. Mat et al. [8] and Aldas et al. [9] studied numerically three-dimensional heat and mass transfer in a metal hydride bed during absorption. Jemni et al. [10] conducted an experimental and numerical study in order to determine the effective thermal conductivity, the equilibrium pressure, and the reaction kinetics of a metal-hydrogen reactor. Demircan et al. [11] examined experimentally the hydrogen absorption in two $LaNi_5$–H_2 reactors. Optimization

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of hydrogen storage in metal hydride beds was investigated by Eustathios et al. [12] who also investigated two novel cooling design options by introducing additional heat exchangers at a concentric inner tube and an annular ring inside the tank. Recently, Kaplan [13] studied the effects of heat transfer mechanisms on the charging process in metal hydride reactors under various charging pressures. Three different cylindrical reactors with the same base dimensions are designed and manufactured. The experimental results showed that the charging of MHVs is mainly heat transfer-dependent and the reactor with better cooling exhibits the fastest charging characteristics. Mellouli et al. [14] designed and manufactured a novel design of MHV equipped with a spiral heat exchanger and mentioned that the charge and the discharge time are significantly reduced. In a previous publication [15], we have measured and modelled the kinetics of hydrogen sorption by $LaNi_5$ and two related pseudobinary compounds. Melnichuk et al. [16] optimized the aluminum fraction of heat transfer fins for a $LaNi_5$ hydrogen storage container, at different absorption times and container diameters. Askri et al. [17] established a model to study the dynamic behavior inside various designs of MHVs. Optimization results indicate that almost 80% improvement of the reaction time can be achieved when the design includes a concentric heat exchanger tube equipped with fins and filled with flowing cooling fluid.

All these studies have shown that the charge/discharge times of MHVs strongly depend on the heat transfer from/to the hydride bed. Thus different attempts to improve the effective thermal conductivity of metal hydride beds have been undertaken integration of copper wire net structure [18] packing in a multilayer waved sheet structure, micro-encapsulated metal hydride compact [19], compacting metal hydride powder with an expanded graphite [20] and insertion of aluminum foam [21]. The integration of the fins represents an important solution to improve the absorption and the desorption processes.

In this paper, we investigate the influence of heat transfer on the reaction kinetic and thermal behavior of two MHVs designs. Experimental results, for the two cases, are compared. In addition, the effect of the variation of the operational parameters on the sorption processes was studied.

2. Experimental set-up

In this study, a novel design of finned spiral heat exchanger was designed and manufactured.

The MHV is manufactured with stainless 316-L; it is composed of two parts, namely:

- A cylindrical body: internal diameter = 80 mm, external diameter = 110 mm, interior height = 110 mm, thickness of the bottom = 15 mm.
- A lid: diameter = 180 mm, thickness = 10 mm

In the basic configuration (Fig. 1), denoted by C1, the heat exchange is carried out through the lateral side of the reactor by means of a heat exchanger. This heat exchanger is implemented by mounting a sealed cylindrical shell outside the reactor and coaxially to it.

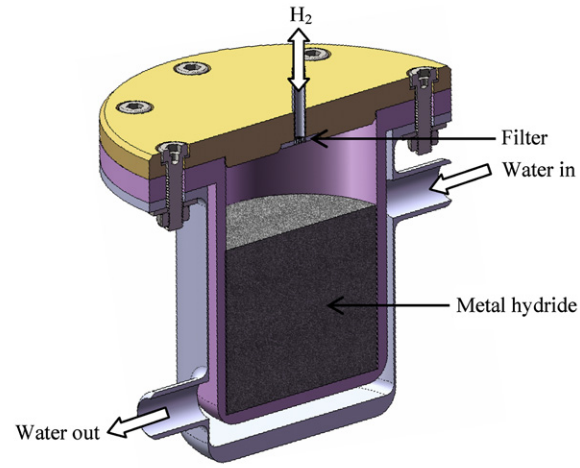

Fig. 1 – Cross section of the basic configuration C1.

Concerning the novel configuration, denoted by C2, the heat exchange takes place within the reaction bed. In order to do so, a finned spiral heat exchanger is immersed in the hydride bed. This thermal exchanger consists of a stainless steel tube of inner and outer diameters of 5 and 6 mm respectively, which is rolled up into a circular helix with 50 mm of diameter and 11 turns with a pitch of 7 mm. Thus, taking into account the outer diameter of the tube and the value of the pitch, we obtain a space between two turns of 1 mm in which we inserted a copper fin coil (Fig. 2).

Both configurations, C1 and C2, have an O-ring sealed flange at one end, at the center of which there is fitted with a porous metal filter. Also, they are well insulated. The schematic of the experimental set-up designed to study the heat and mass transfer characteristics of the three MHSV is shown in Fig. 3.

Set-up mainly consists of a hydride reactor, two reference cylinders interconnected via valves allowing the variation of the reference volume (7600 Cm^3), a vacuum pump to evacuate the reactor and the reference cylinders before, absorption and desorption experiments respectively, a thermostatic water bath (0–100 °C), a water pump, a data acquisition system and a tank of 99.99% pure hydrogen gas. The hydriding and dehydriding processes are monitored with temperature and pressure measurement via three thermocouples positioned at three locations within the hydride bed and with a pressure sensor installed between the reference cylinders and the hydride reactor. The temperature and the pressure readings are recorded on a computer for further processing and interpretation.

3. Experimental procedure

The material selected to fill both configurations is the $LaNi_5$ alloy which exhibits excellent volumetric density, easy activation and moderate plateau pressures. The reaction volume is loaded with 1 kg of mechanically milled $LaNi_5$. Before

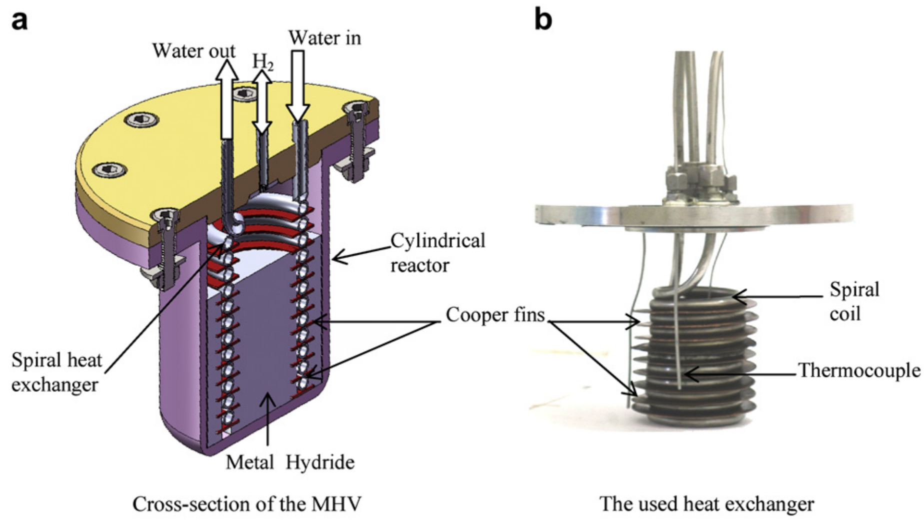

Fig. 2 – MHV based on spiral heat exchanger with fins.

starting the study of the sorption phenomena, the activation of the alloy is executed. It consists of the repetition of an absorption/desorption cycles (25 cycles of absorption/desorption) until the maximum hydrogen storage capacity (14 w%) of the metal is reached.

The vacuum in the reactor and the tanks is ensured by a vacuum pump. Then, thermostated water flow through the heat exchanger to bring the reactor to the temperature value which will be the initial and final state of the absorption test. Once the reference volume in fixed, hydrogen is loaded at the desired pressure. The reactor and the reference volume are then interconnected and the data acquisition, which includes hydrogen dynamic pressure and temperatures of hydride bed and water outlet, is executed. The absorption experience continues until steady state is reached.

The experimental procedure for desorption starts at the end of an absorption test. Reference volume is emptied via the vacuum pump, then connected to the reactor and, simultaneously, starting the circulation of the thermostated water at the desired temperature until reaching a steady state. After that we begin the data acquisition.

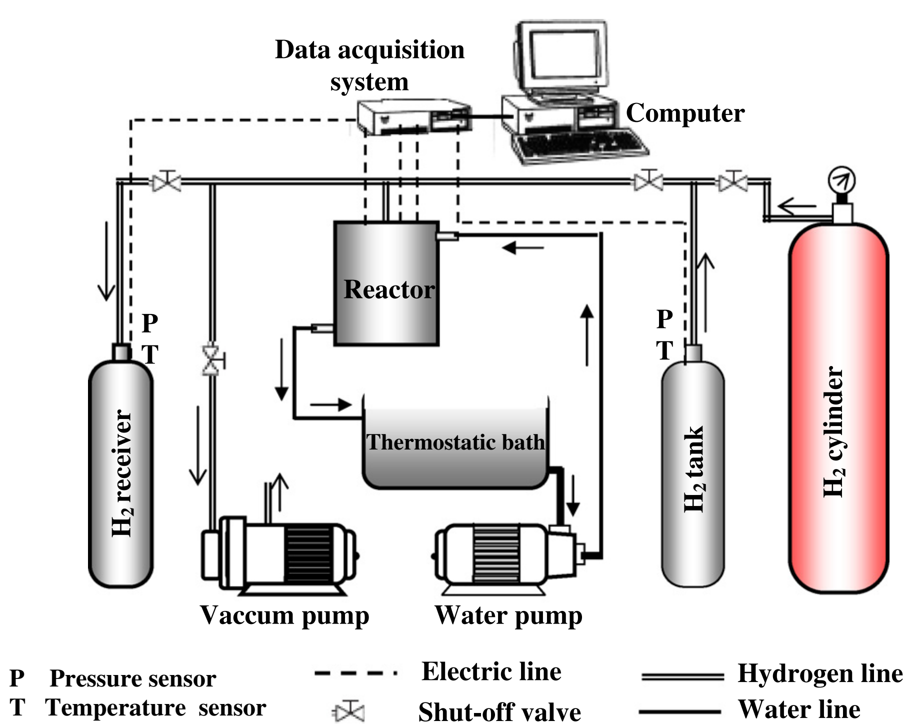

Fig. 3 – Synoptic scheme of the experimental set-up.

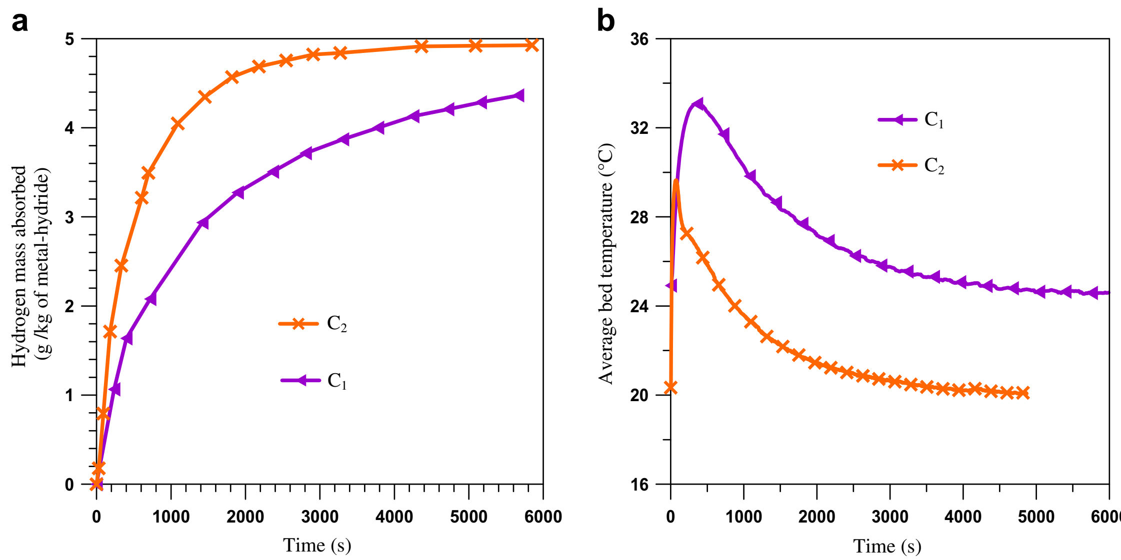

Fig. 4 – Evolution of the absorbed hydrogen mass and the average bed temperature for both configurations C1 and C2.

4. Experimental results

Fig. 4 displays the progress of the absorbed hydrogen mass and the average bed temperature for the two configurations C1 and C2.

According to Fig. 4a, the hydrogen absorption kinetic is improved for the novel configuration and more hydrogen was stored during the experience. From Fig. 4a we note that the time at which 4 g of hydrogen has been stored is 1000 s for the C2 configuration and 3700 s for C1 configuration.

It's clear from Fig. 4b, that for C2 configuration the temperature remains permanently below that of the C1 configuration; this explains the improvement of the absorption kinetic observed for the new configuration. Indeed, due to the exothermic character of the absorption reaction, the charging of MHV is mainly heat transfer-dependent and the reactor with better cooling exhibits the fastest charging characteristics.

Actually, it's well known that one of the main factors governing the absorption reaction is the driving force for mass transfer which is defined as the difference between the hydrogen supply pressure and the equilibrium pressure Fig. 5 shows the evolution of the equilibrium pressure of the hydride bed for the two configurations.

At the beginning of the hydriding reaction, the absorption rate depends to the limited heat conductivity of the hydride bed; subsequently most of the released energy due to hydrogen absorption heats the hydride bed itself which is reflected by a sudden increase of the reactor average temperature (Fig. 4b).

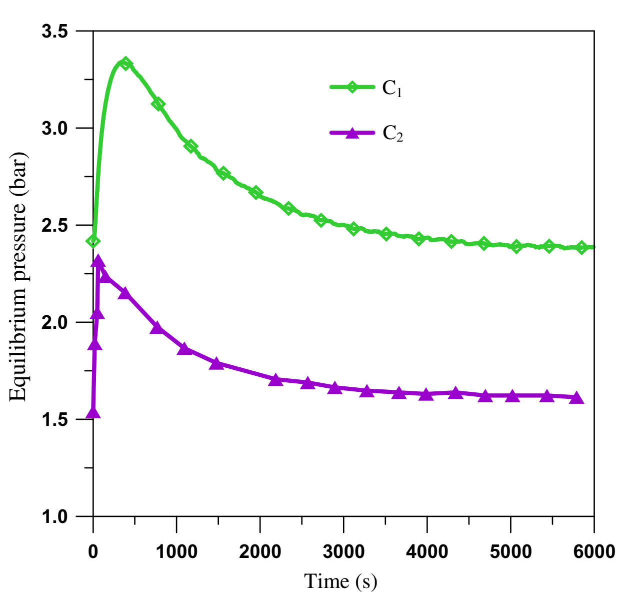

Fig. 5 – Evolution of the equilibrium pressure for both configurations C1 and C2.

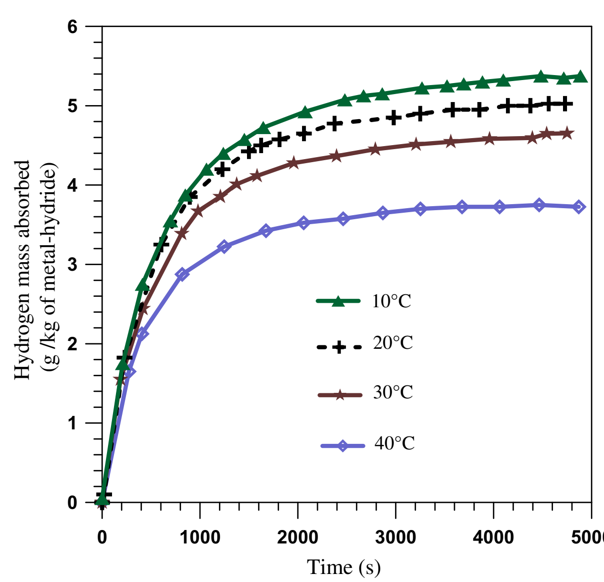

Fig. 6 – Effect of the cooling water temperature on the kinetics of absorption.

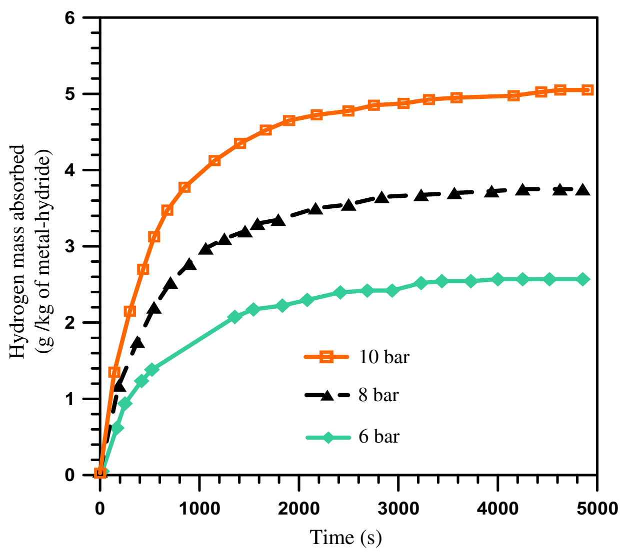

Fig. 7 – Effect of the initial hydrogen pressure on the absorption kinetic.

Also, this has an effect of increasing the equilibrium pressure, therefore reducing the driving force for mass transfer. It is evident from Fig. 5 that the configuration C2 has the greatest driving force for mass transfer during absorption.

Fig. 6 shows the evolution of the absorbed hydrogen mass in the case of the C2 reactor at different temperatures of the thermostated water. By lowering the cooling temperature, the temperature gradient between the hydride bed and water increases. This helps to cool more effectively the sample of $LaNi_5$, therefore, accelerate the kinetic of the exothermic absorption reaction.

By setting coolant water at 20 °C, absorption pressure is $P_0 = 10$ bars and flow mass of water is 13 g/s, we can evaluate the effect of the pressure on the absorption phenomenon. Fig. 7 shows the influence of the initial hydrogen pressure at constant reference volume on the absorption kinetic. For a higher value of initial pressure, there would be more hydrogen mass available in the system, thus a higher charge of the hydride would be possible before the pressure of the experimental systems drops. In this case, the supply pressure is responsible for raising the driving force for mass transfer since the equilibrium pressure is set by the initial experimental conditions. Thus, increasing the initial hydrogen pressure, simultaneously, speeds up the absorption reaction and stores more hydrogen in the $LaNi_5$ sample.

If we maintain constant the temperature and flow of the coolant and the initial hydrogen supply pressure, we can assess the influence of the reference volume on the absorption kinetic. Fig. 8 summarizes the different results.

It is clear that for a larger reference volume, the absorption kinetic is improved. This observation implies that the driving force for mass transfer would be affected by the variation of the reference volume. Given the initial experimental conditions, the amplitude of the driving force for mass transfer, according to its definition, remains the same for each experiment at the beginning of the experiment. However, this amplitude decreases less rapidly if we increase the reference volume since the applied hydrogen pressure decreases more slowly (Fig. 8).

Also according to Fig. 8a, the total hydrogen mass stored increases; this is legitimate because by increasing the reference volume at a constant initial gas pressure, we offer more hydrogen mass to the kilogram of $LaNi_5$ to absorb.

The effect of varying the reference volume on the average temperature of the reactor C2 is shown on Fig. 9.

The maximum average temperature reached at the beginning of the reaction of absorption is not significantly affected by the variation of the reference volume. However the

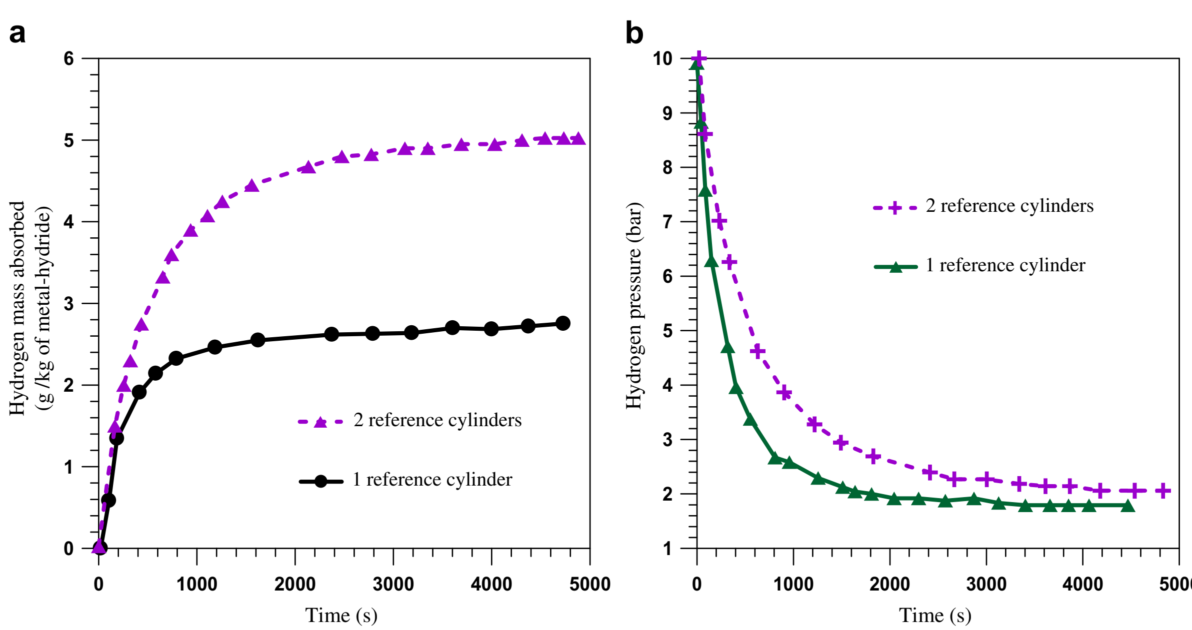

Fig. 8 – Effect of the reference volume on (a) the absorption kinetic and on (b) the evolution of the hydrogen pressure.

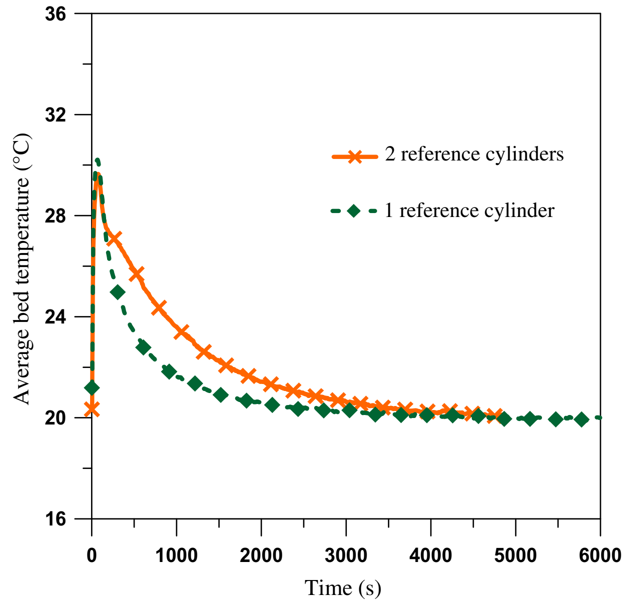

Fig. 9 – Effect of the reference volume on the average temperature of the hydride bed.

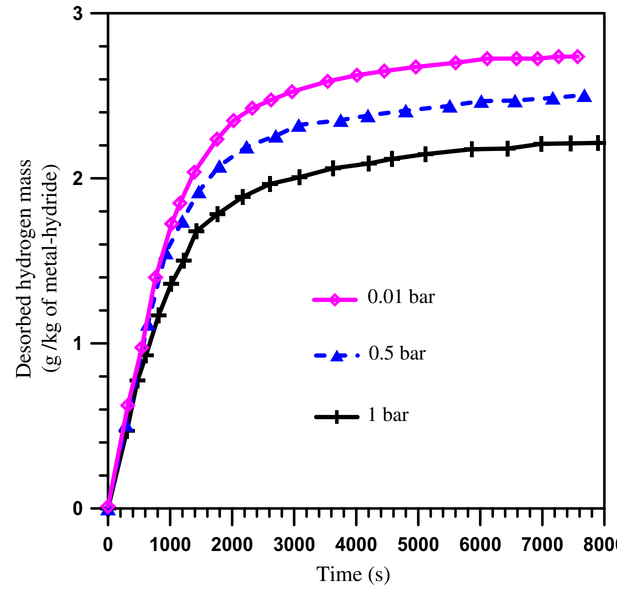

Fig. 11 – Effect of initial drain pressure on the desorption kinetic.

return of the average temperature to its initial value (that of the cooling water: 20 °C) is delayed more as the reference volume is enlarged. In fact, since the pressure decrease is slower for the 2 reference cylinders case, the rate of the hydriding reaction is maintained relatively high, therefore more mass of hydrogen is absorbed and more heat of reaction is produced.

Desorption process has responses to the changes of the pressure and the temperature similarly to the absorption reaction. Thus, the same interpretations and observations are valid for the desorption analysis. Heating is an important issue for the desorption reaction owing its endothermic character. Fig. 10 shows that the more heat is provided to the hydride bed, the better is the desorption kinetic. If we decrease the initial pressure in the reference volume, the desorption reaction increases and the mass of desorbed hydrogen increases (Fig. 11).

If we increase the reference volume, it is evident from Fig. 12 that desorption kinetics is enhanced and the mass of recovered hydrogen is greater because the driving force of mass transfer is maintained longer for a bigger drain tank.

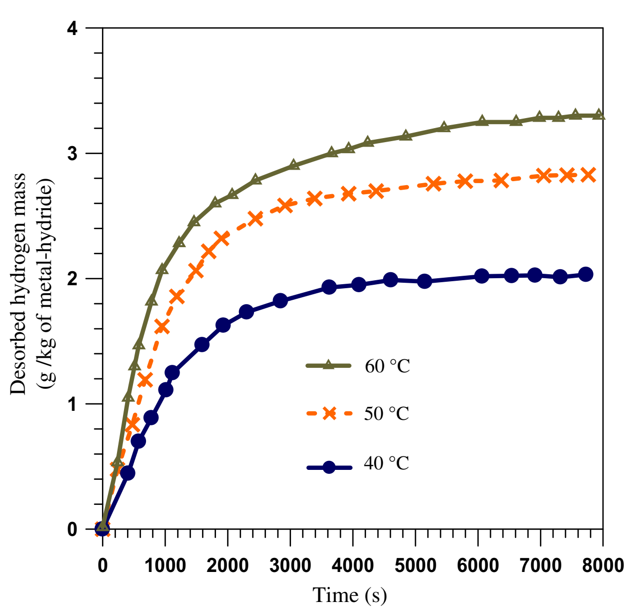

Fig. 10 – Effect of heating temperature on the desorption kinetic.

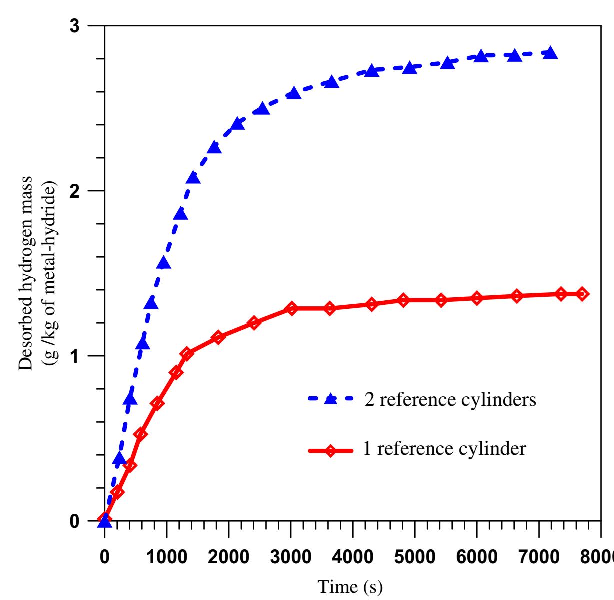

Fig. 12 – Effect of draining volume on the desorption kinetics.

5. Conclusion

In this study, an experimental set-up was designed to study the effect on the hydrogen sorption performance of operational parameters for a metal hydride vessel with a finned spiral-type heat exchanger. The experimental results show that the absorption/desorption times of the MHV are considerably reduced due to the integration of the finned heat exchange system. In addition, the effect of different parameters (flow mass and temperature of the cooling/heating fluid) has been discussed and obtained results show that a good choice of these parameters is important for the control of the charge/discharge times.

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